Disentangling the Black Box: Interpretable Protein-Ligand Docking via Sparse Autoencoders

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Abstract

Protein-ligand docking pipelines based on variational autoencoders (VAEs) achieve strong predictive performance but produce opaque latent representations, while the large number of candidate poses generated per complex imposes substantial computational cost as each undergoes expensive energy minimization and refinement regardless of quality. Physics-informed generative pipelines like the Friesner Group's IFD-ML pipeline ground predictions in thermodynamic principles, making learned features more likely to be physically meaningful and interpretability more tractable and valuable than in end-to-end models. We apply sparse autoencoders (SAEs) to the VAE latent space of the IFD-ML pipeline, training on 435,069 poses across 36 protein systems. At a 4Å threshold, SAE-derived features filter 16.5% of poor poses prior to refinement while retaining 91.1% of native-like conformations, outperforming the pipeline's own energy score. Fraction-of-variance-explained (FVE) analysis shows the top SAE retains 62% of latent variance while still supporting a high-precision filter. Sparse features activate exclusively on failed docking attempts, each corresponding to a distinct structural failure mode—an interpretable window into what generative models learn about binding physics. These results show SAEs can serve simultaneously as practical computational filters and interpretability tools for building trustworthy drug-discovery pipelines.

1 Introduction

Protein-ligand binding governs essentially all biological processes and underlies the majority of small-molecule drug discovery [Dhakal et al., 2022, Sim et al., 2025]. Predicting how a ligand binds a flexible protein target, the induced fit docking (IFD) problem [Sherman et al., 2006], remains one of computational biochemistry's hardest tasks, addressed by physics-based pipelines such as IFD-MD [Miller et al., 2021]. Most end-to-end machine learning (ML) approaches, such as DeepDTA and its graph-based successors, treat binding as a sequence-to-structure mapping without explicitly modeling atomic interactions [Wu et al., 2025], and are further limited by scarce training data, poor interpretability, and weak generalization to novel proteins or ligands [Lam and Katritch, 2025, Jiang et al., 2026].

Schrödinger and the Friesner Group's IFD-ML pipeline combines physics-based simulation with variational autoencoder (VAE) and diffusion generative models to sample conformational space, achieving strong performance in cross-docking benchmarks similar to Mansoor et al. [2024] (Konstantinovsky et al., in preparation). Because it relies on a VAE, each candidate pose is represented by a fixed-dimensional continuous latent vector—the kind of representation a sparse autoencoder (SAE) is designed to decompose into interpretable features (Section 2.2). This creates two problems. First, the pipeline generates thousands of candidate poses per complex, each undergoing expensive Prime energy minimization [Li et al., 2011] and algorithmic refinement regardless of quality—a bottleneck at scale [Wang et al., 2019]. Second, the VAE's latent vectors are opaque: it is unclear which directions encode binding geometry, energetic favorability, or structural quality, and understanding

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

why a model scores a pose highly matters as much as the score itself [Qadri et al., 2025]. This paper addresses both: a practical filter that removes likely-poor poses before refinement, and an interpretability method that opens up what the latent space has learned.

Sparse autoencoders (SAEs) decompose opaque neural representations into interpretable, monosemantic features [Templeton et al., 2024, Gao et al., 2024]. Applied to protein language models, SAEs recover biologically meaningful features corresponding to secondary structure, binding sites, and evolutionary conservation [Adams et al., 2025, Simon and Zou, 2025]. Physics-informed pipelines like IFD-ML are a compelling target: because the latent space is constrained by thermodynamic priors and atomic interactions, learned features are more likely to correspond to real physical quantities—energy terms, geometric constraints, binding physics—than in end-to-end models [Li et al., 2026, Wu et al., 2024]. SAEs have been applied to structure-generation models such as RFdiffusion (FoldSAE) [Zarzecki et al., 2025] and protein folding representations (ESMFold/ESM2-3B) [Parsan et al., 2025], but to our knowledge not yet to a physics-informed, molecular-docking generative model.

This work applies TopK SAEs to the VAE latent space of the IFD-ML pipeline to serve both goals at once. Concretely, we make three contributions: (1) a practical pre-refinement filter that flags 16.5% of poses for removal while retaining 91.1% of native-like conformations; (2) evidence that the pipeline's own energy score cannot substitute for this filter, structurally unable to reach the SAE's operating points; (3) sparse features that isolate distinct, physically interpretable failure modes, from severe rotational misorientation to milder translational displacement, in what generative models learn about binding physics.

2 Methods

2.1 Dataset

We compiled a dataset of 435,069 molecular docking poses across 36 protein-ligand systems generated by the IFD-ML pipeline, split by case (not pose) to prevent leakage. Each pose consists of a 30-dimensional VAE latent vector, ligand RMSD to the native structure (excluding hydrogen atoms), and a free energy prediction from physics-based scoring. We use two RMSD-based pose-quality definitions: native-like at RMSD ≤ 2Å (12.3% native-like, following Shim et al. [2022]), and good at RMSD ≤ 4Å (36.6% native-like) for the conservative filtering task. Splits are approximately 70% train, 15% validation, 15% test; normalization uses the training split only, and feature/threshold/combo selection for filtering (Section 2.4) draws on pooled train+val, with test reserved for a single final evaluation.

2.2 TopK Sparse Autoencoder

We train TopK SAEs with $4\times$ expansion (30D input $\rightarrow$ 120D hidden $\rightarrow$ 30D reconstruction) following Adams et al. [2025]; the TopK operation retains the K largest activations and zeros the rest, with reconstruction loss as the training objective. The input is mean-centered via a learned pre-encoder bias, initialized to the training-set mean of the normalized activations and subtracted before encoding, following Simon et al. [2026]'s dead-feature mitigation. We trained SAEs across $K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds per K, for up to 100 epochs using Adam (lr $= 10^{-3}$, batch size 256), selecting the best run per K by validation loss; we report results for $K \in \{3, 8, 15\}$ in the main text (the full sweep across all seven K values follows similar trends but is omitted here for space), selecting the operating K per analysis (Section 3).

2.3 Pose Quality Classification

To assess whether SAE features provide utility beyond raw latents, we trained logistic regression classifiers on (1) 30D normalized VAE latents (L2 penalty, class-balanced) and (2) 120D sparse SAE activations for $K \in \{3, 8, 15, 20, 30\}$ (L1 penalty, class-balanced), fit on train and evaluated on test. auPR is our primary metric, since it handles class imbalance better than ROC-AUC, which we also report for comparison with prior work.

2.4 Threshold-Based Pose Filtering

The paper's central practical result is a pre-refinement filter, built using a three-way train/select/test split so reported performance cannot be inflated by threshold or feature selection on the test set. For each of the 120 features per SAE, we fit an F1-maximizing activation cutoff and rank candidates by precision on a pooled train+val selection set, requiring recall > 0.02 to exclude degenerate near-zero-recall picks. We take the top 16 candidates, then search all subset unions—evaluated on the same selection set—for the union maximizing recall subject to retaining $\geq$80% of native-like poses; the winning union is applied exactly once to test, which is never touched during fitting or selection. To confirm that precision derives from the SAE decomposition rather than raw latent geometry, we apply the identical procedure to 500 random unit directions and all 60 signed raw latent axes. Finally, as a physics-based baseline, we apply the same discipline directly to the IFD-ML pipeline's pre-minimization energy score, accounting for its sentinel value (10,000 kcal/mol)—assigned when the ligand is not in the basin that refinement operates in, or the pose is otherwise too structurally poor to score meaningfully.

3 Results

3.1 Reconstruction, Classification, and Sparsity

Does decomposing the latent space into sparse features cost any of the predictive signal present in the raw, dense latents? Table 1 reports auPR for bad-pose classification, FVE, and dead-feature rate, all on the test set. Raw latents match or exceed every SAE K on auPR at both thresholds—the sparse decomposition provides no consistent predictive lift, so its value lies in filtering (Section 3.2) and interpretability (Section 3.4) rather than classification. FVE grows monotonically with K without a matching auPR gain, and dead-feature rate spikes to 25.8% at K=15 despite the mean-centering mitigation from Section 2.2 [Simon et al., 2026].

Table 1: auPR, FVE, and dead-feature rate, raw latents vs. SAE K (test set).

Rep.	auPR (4Å)	auPR (2Å)	FVE	Dead %
Raw latents	0.770	0.855	—	—
SAE K=3	0.709	0.858	33.7%	2.5%
SAE K=8	0.719	0.821	62.0%	1.7%
SAE K=15	0.750	0.846	90.0%	25.8%

3.2 Practical Filtering

We now evaluate the pre-refinement filter from Section 2.4. Table 2 reports the combo filter for $K \in \{3, 8, 15\}$ at both thresholds, alongside the raw-latent-geometry baselines (Section 2.4). K=8 and K=15 are the reproducible headline operating points, at 4Å and 2Å respectively (K=3's combo cleared the 80% good_kept bar on the selection set at 85.2% but dropped to 77.5% on held-out test—a genuine generalization failure rather than a threshold artifact): K=8 flags 16.5% of test poses while retaining 91.1% of native-like conformations (precision 0.803); K=15 flags 15.7% while retaining 93.0% (precision 0.935). Recall is modest at both (0.21, 0.17)—a deliberate precision-first design choice, since retaining native-like poses matters more than catching every poor one for a pre-refinement filter.

Neither raw-latent-geometry baseline family reaches the good_kept $\geq$ 80% target at either threshold (0 of 65,535 candidate-union combinations clear it), so Table 2 reports each baseline's closest achievable point instead—well below what K=8/K=15 achieve while flagging fewer poses.

3.3 Comparison to the IFD-ML Pipeline's Energy Score

Applying the same selection discipline (Section 2.4) to the pipeline's pre-minimization energy score, 38.4% of test poses at both thresholds are tied at the sentinel value (energy = 10,000 kcal/mol)—occurring when the ligand starts outside the basin that refinement operates in, or the pose is otherwise unsalvageable. Because this is a tied point mass, no energy threshold can flag fewer than 38.4% of poses regardless of tuning—both SAE headline operating points fall below this floor and so are structurally unreachable, not merely outperformed. At its most conservative achievable point (the

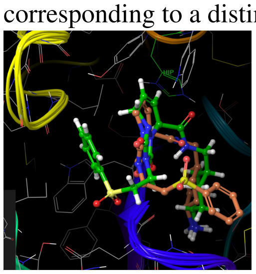
Table 2: Filtering performance: SAE $K \in \{3, 8, 15\}$ vs. raw-latent-geometry baselines, both thresholds, held-out test set.

Threshold	Source	% Flagged	Precision	Recall	Good Kept
4Å	SAE K=3	27.3	0.698	0.301	77.5%
4Å	SAE K=8 (headline)	16.5	0.803	0.210	91.1%
4Å	SAE K=15	12.3	0.879	0.171	95.9%
4Å	Random directions (best)	33.7	0.605	0.321	63.6%
4Å	Raw latent axes (best)	27.8	0.727	0.319	79.2%
2Å	SAE K=3	38.1	0.791	0.353	45.8%
2Å	SAE K=8	19.7	0.884	0.204	84.4%
2Å	SAE K=15 (headline)	15.7	0.935	0.172	93.0%
2Å	Random directions (best)	33.7	0.824	0.325	59.7%
2Å	Raw latent axes (best)	27.8	0.857	0.280	72.8%

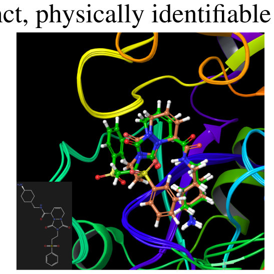
sentinel-only filter, 38.4% flagged), energy reaches 78.9% good_kept at 4Å and 78.2% at 2Å—both well below the SAE filters despite flagging more than double the poses. An unconstrained best-F1 energy threshold performs even worse, degenerating to flagging 100% of poses.

3.4 Interpretable Features

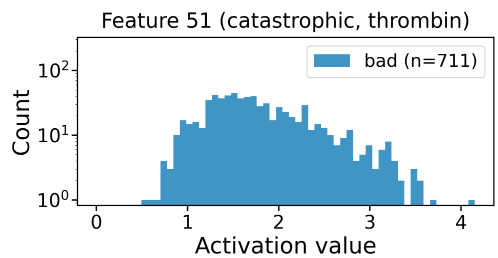
The SAE's sparse features are not just useful for filtering—they are individually interpretable, each corresponding to a distinct, physically identifiable failure mode.



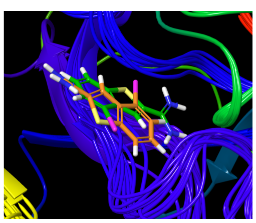
(a) Pose 2384 (RMSD 5.6Å)



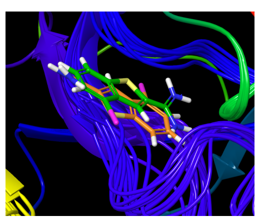
(b) Pose 4556 (RMSD 7.2Å)



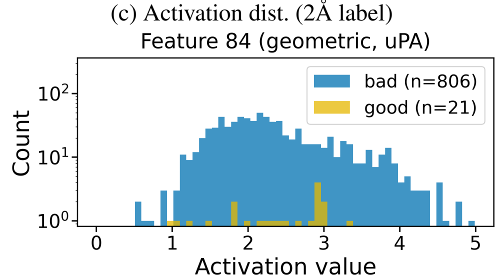
(c) Activation dist. (2Å label)



(d) Pose 11084 (RMSD 5.7Å)



(e) Pose 11377 (RMSD 5.0Å)



(f) Activation dist. (2Å label)

Figure 1: Feature 51 (top, thrombin) and Feature 84 (bottom, uPA), both members of the deployed K=8 4Å combo filter (case details and statistics in text). Left/middle: top-activating test poses, docked pose (orange) vs. reference (green); the colored backbone traces in Feature 51's poses reflect the pipeline's induced-fit receptor-loop sampling, present in both good and bad poses for this case (see below). Right: activation distribution (active poses only, log-scale count) split by the 2Å native-pose criterion.

Table 2's K=8 combo filter is a union of 16 candidate features (Section 2.4); we highlight two with sharply contrasting structural signatures for detailed interpretation. Feature 51 fires on 711 test poses, 99.3% of which come from a single thrombin case pair [Di Cera et al., 1995] (PDB 1AE8→1D9I [De Simone et al., 1997, Krishnan et al., 2000], 706/711; a second thrombin pair accounts for the remaining 5), at precision 0.809 with a sentinel rate of 86.4%. This is not indiscriminate case detection: the feature fires on just 9.5% of that case's poses and 10.8% of its bad poses. Under the stricter, standard 2Å native-pose criterion, precision is perfect: all 711 flagged poses are non-native (0 false positives). Its highest-activating poses (Figure 1) show a case-specific rotational failure: the ligand's sterically bulky substituent, which should point toward Leu99, instead points toward

Ala190—both residues lining the S1/S2 subsites adjacent to the catalytic triad [Bode et al., 1989]. This misorientation is severe enough that most flagged poses fall outside any refinable basin.

Feature 84 tells a different story. It fires on 827 poses, 94.3% concentrated in a single urokinase (uPA) case pair [Spraggon et al., 1995] (PDB 2VIO→1C5W [Frederickson et al., 2008, Katz et al., 2000], 780/827)—the same catalytic serine-protease fold family as thrombin—at precision 0.911 with a sentinel rate of 28.5%, *below* the population baseline; at the 2Å native-pose criterion, precision rises to 0.975 (806/827). Its highest-activating poses show the ligand correctly oriented but translationally displaced within the binding pocket; energies converge to real, physically stable values (clustered around −9,300 to −9,700 kcal/mol) rather than the sentinel value. On the 20 most confidently native poses for this case (RMSD <0.4Å), feature 84's activation is exactly 0.0000—the feature is silent on good poses rather than weakly active everywhere.

Together the two features illustrate a rotational-vs-translational split in the geometric errors the SAE decomposes into distinct, monosemantic features: severe rotational misorientation that falls outside any refinable basin (Feature 51), and milder translational displacement that minimization tolerates (Feature 84)—consistent with the monosemanticity hypothesis that sparse features encode distinct failure phenomena rather than one distributed correlate of pose quality [Templeton et al., 2024, Adams et al., 2025, Clark et al., 2026, Bricken et al., 2023].

Case-specific concentration is not unique to Features 51 and 84: across all 16 candidates in the K=8 combo (Table 3, test set), 5 concentrate ≥90% of firings in a single case pair and 12 concentrate ≥50%, spanning four structurally and functionally distinct targets—the two proteases already discussed, a phosphatase (PTP1B), and an ionotropic glutamate receptor (GluR5)—rather than one fold family. A mirror-image search (features preferring native-like poses) found nothing comparably precise or concentrated, and the induced-fit receptor-loop flexibility visible in Feature 51's poses (Figure 1) is not diagnostic—the same spread appears in the case's most confidently native poses.

Table 3: Case-concentration of the K=8 combo filter's 16 candidate features, by dominant protein target (test set).

Target	# features	Mean dom. case %	Max dom. case %
GluR5 (ion channel)	3	81.4%	100.0%
Thrombin (protease)	8	62.2%	99.3%
PTP1B (phosphatase)	1	98.3%	98.3%
uPA (protease)	4	74.2%	95.5%

4 Conclusion

We have shown that sparse autoencoders can decompose the latent space of a physics-informed docking pipeline into interpretable, case-specific features—high-precision detectors for distinct, rare failure modes. The deployed combo filters remove a substantial share of poor poses while retaining the large majority of native-like conformations at both thresholds (Section 3.2), beating both a raw-latent-geometry baseline and the pipeline's own energy score, structurally unable to reach these operating points (Section 3.3). SAEs offer a path toward docking pipelines that are not only predictive but mechanistically understood—systems whose failures can be diagnosed, explained, and, eventually, corrected.

That case-specific concentration extends across the K=8 combo's 16 features (Section 3.4) suggests the SAE learns protein-specific failure signatures a global classifier would dilute—these filters are most valuable when the target family is known, complementary to case-agnostic scoring. Recall is modest at both headline operating points, a deliberate precision-first choice rather than a detectability ceiling, and generalization beyond the 36 systems studied here requires further validation. Understanding a generative pipeline's failures is one step closer to making physics-informed drug discovery interpretable.

Broader Impact This work reduces computational cost and improves interpretability by filtering low-quality poses before refinement and surfacing failure modes the model has not learned. Filtering existing poses, rather than generating new molecules, does not lower the barrier to designing toxic compounds—the field's primary dual-use concern. A more realistic risk is uneven performance: our filters are case-specific (Section 3.4), so precision may not transfer beyond these 36 systems, and over-reliance could deprioritize rarer disease targets.

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Justification: Section 2.1 specifies the case-based 70/15/15 train/validation/test split and normalization discipline; Section 2.2 specifies the TopK SAE architecture, K sweep, seeds, optimizer (Adam, lr 10^{-3}), batch size, and epoch budget; Section 2.4 specifies the feature-selection and combo-filter procedure in full.

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Justification: SAE training was run on an internal GPU cluster (16 nodes, 8 GPUs each: nodes gpu001–gpu008 with NVIDIA RTX A4000, 16GB VRAM; gpu009–gpu016 with RTX A4500, 20GB VRAM), with each of the 35 training runs ($K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds) using a single GPU. Given the small model size (30D→120D→30D), per-run compute is modest; however, exact wall-clock time and total compute were not logged and are not reported.

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